

Risk Documentation

Smart House System

Project's Github repo link:

<https://github.com/DanielJ0131/interactive-house>

Revision History

Date	Version	Description	Author
2026-02-08	1.0	Initial potential risks for the Smart House System	Daniel Marcarini
2026-02-24	1.1	AI Risk Extension (R10-R12)	Md Abu Taher
		AI Risk Extension (R13-R14)	Anitta Antony
2026-02-28	1.1	Smart Home code	Daniel Marcarini
2026-02-28	1.1	Speech Risk (R15)	Daniel Holderik
2026-03-28	1.2	Updated with Web and Firebase	Md Abu Taher Anitta Antony
2026-03-28	1.2	Testing	Daniel Marcarini
2026-04-22	1.3	Added measurable AI response-time thresholds	Anitta Antony Md Abu Taher
2026-05-08	1.4	Updated AI risk management status and mitigation plans	Anitta Antony Md Abu Taher
2026-05-08	1.5	Updated testing risk mitigation after adding Jest automated tests and Istanbul coverage for web authentication and Firebase configuration	Daniel Marcarini
2026-05-21	1.6	Updated R15	Daniel Holderik
2026-05-23	Final	Updated final testing status and project completion evidence	Daniel Marcarini

Risk List

Risk Description	Priority	Progress
R1. Integration Complexity Between System Components	High	Managed
R2. Scalability and Performance Limitations	Medium-High	Managed

R3. Incomplete or Over-Ambitious Innovation Scope	High	Managed
R4. Unclear Requirements or Changing Requirements	Medium	Managed
R5. Unauthorized Access or Security Vulnerabilities	High	Managed
R6. Data Privacy Concerns	Medium-High	Managed
R7. Uneven Work Distribution	Medium	Managed
R8. Time Constraints and Deadline Pressure	Critical	Managed
R9. Insufficient Testing	High	Managed
R10. AI misinterpretation of User commands	High	Managed
R11. Delay in AI Response Time	Medium- High	Managed
R12. Over-Reliance on AI Decision Logic	Medium	Managed
R13. AI Service Dependency and External AI Service Failure	High	Managed
R14. Incorrect AI response in safety-critical Scenarios	High	Managed
R15. Dependency on external speech recognition library	High	
R16. insufficient verification of web application and device-control integration	High	Managed

Risk Handling Plans

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Risks – SG4 Innovation

1 Introduction

This section identifies and evaluates the key risks associated with the SmartHouse project, particularly those related to innovation, system complexity, and technical integration. Since SG4 focuses on innovation and refinement, several technical and project-related uncertainties must be considered to ensure successful implementation and delivery.

Each risk is evaluated in terms of:

- **Description**
- **Likelihood** (Low / Medium / High)
- **Impact** (Low / Medium / High)
- **Mitigation Strategy**

Technical Risks

Risk 1: Integration Complexity Between System Components

Description:

The SmartHouse system consists of multiple interacting components (e.g., sensors, controllers, user interface, backend logic, possible IoT simulation). Integration between these modules may introduce unexpected behavior, compatibility issues, or architectural inconsistencies.

Likelihood: Medium

Impact: High

Mitigation Strategy:

- Use a modular architecture with clearly defined interfaces.
- Apply early integration testing instead of postponing integration until late stages.
- Follow consistent design principles and coding standards.
- Conduct peer code reviews.

Risk 2: Scalability and Performance Limitations

Description:

Innovative features (e.g., automation rules, real-time monitoring, event handling) may increase system complexity. If not properly designed, the system may experience performance degradation as the number of devices or rules increases.

Likelihood: Medium

Impact: Medium to High

Mitigation Strategy:

- Use efficient data structures and algorithms.
- Perform early performance testing with simulated load.
- Avoid unnecessary polling or blocking operations.
- Refactor performance-critical sections when identified.

Risk 3: Incomplete or Over-Ambitious Innovation Scope

Description:

Since SG4 emphasizes innovation, there is a risk that the team attempts to implement overly complex or experimental features that exceed the time or technical capacity of the group.

Likelihood: Medium

Impact: High

Mitigation Strategy:

- Prioritize core functionality before innovative extensions.
- Define a Minimum Viable Product (MVP).
- Time-box experimental features.
- Regularly evaluate progress against deadlines.

Risk 4: Unclear Requirements or Changing Requirements

Description:

Smart house systems often involve evolving user expectations. Changes in requirements during development may cause rework and architectural instability.

Likelihood: Medium

Impact: Medium

Mitigation Strategy:

- Clearly document functional and non-functional requirements.

- Use version control for requirement changes.
- Apply incremental development with frequent feedback.
- Maintain traceability between requirements and implementation.

Security and Privacy Risks

Risk 5: Unauthorized Access or Security Vulnerabilities

Description:

A smart house system may simulate or implement authentication, device control, and user data handling. Weak security design may lead to vulnerabilities such as unauthorized access or manipulation of devices.

Likelihood: Medium

Impact: High

Mitigation Strategy:

- Implement proper authentication and authorization mechanisms.
- Validate all user inputs.
- Avoid hard-coded credentials.
- Follow secure coding practices.

Risk 6: Data Privacy Concerns

Description:

If the system stores user preferences, logs, or sensor data, there is a risk of mishandling sensitive information.

Likelihood: Low to Medium

Impact: High

Mitigation Strategy:

- Minimize stored personal data.
- Apply encryption if sensitive data is stored.
- Clearly define data storage policies.
- Restrict access to stored data.

Project Management Risks

Risk 7: Uneven Work Distribution

Description:

In group-based software engineering projects, some members may become overloaded while others contribute less, affecting quality and deadlines.

Likelihood: Medium

Impact: Medium

Mitigation Strategy:

- Clearly define responsibilities.
- Use task tracking tools.
- Hold regular team meetings.
- Ensure transparency in workload distribution.

Risk 8: Time Constraints and Deadline Pressure**Description:**

Innovation often requires experimentation and iteration, which may conflict with fixed academic deadlines.

Likelihood: High

Impact: High

Mitigation Strategy:

- Create a detailed project timeline.
- Prioritize essential features.
- Reserve buffer time for testing and debugging.
- Monitor progress weekly.

Quality Risks**Risk 9: Insufficient Testing****Description:**

Due to focus on innovation and feature development, testing may be deprioritized, resulting in unstable functionality.

Likelihood: Medium

Impact: High

Mitigation Strategy:

- Implement unit testing for core components.
- Perform integration testing before delivery.
- Use test cases derived from requirements.
- Automate testing where possible.

Current Mitigation Status:

An initial automated test suite has been added to reduce the risk of insufficient frontend testing. The web application now uses Jest and React Testing Library to test authentication-related user flows and Firebase configuration. The current automated suite includes 14 passing tests across 4 test suites.

The tests cover login rendering, login validation, successful login behavior, failed login behavior, signup rendering, signup validation, successful signup behavior, duplicate email handling, Firebase Auth configuration, and Firestore configuration.

Istanbul coverage reporting has also been enabled through Jest to make it easier to identify which parts of the source code are currently covered by automated tests.

Remaining risk:

The risk is reduced but not fully removed. Main smart house features such as hub device control, Firestore live synchronization, emergency flow, music page, AI page, and speech-related behavior still require additional automated or manual testing.

AI Risks

The AI-related risks are managed by identifying possible failures in AI command interpretation, response time, external AI service dependency, safety-critical behaviour, and speech recognition support. The mitigation strategy focuses on keeping safety-critical actions outside the AI module, giving visible feedback during AI processing, providing fallback behaviour when AI or speech input is unavailable, and limiting AI control to supported device actions. Risks marked as managed have mitigation actions already included in the design or implementation. Risks marked as partially managed still depend on external platform support, browser compatibility, or future testing.

Final Mitigation Status

The risk of insufficient testing was reduced by implementing an automated Jest test suite for the web application. The final suite contains 49 passing tests across 9 test suites.

The tests cover authentication, Firebase configuration, shared components, static pages, guest hub, emergency flow, hub device controls, and music behavior. Istanbul coverage reporting was also generated to support visibility into tested areas.

Remaining testing risk is limited mainly to future AI and voice edge cases, browser differences, and real-device integration scenarios.

Risk 10: AI misinterpretation of User commands

Description:

Since the system uses natural language input, the AI may misunderstand unclear or ambiguous user commands. This may result in an incorrect response or an unintended device-control action, such as controlling the wrong device or applying the wrong action.

Likelihood: Medium

Impact: Medium to High

Mitigation Strategy:

- Use structured prompting to guide the AI response.
- Limit AI-controlled actions to supported devices and supported commands.
- Display clear feedback to the user after processing a command.
- Treat confirmation before physical or safety-related actions as a future improvement.

Risk 11: Delay in AI Response Time

Description:

AI processing depends on communication with an external AI service, which may introduce delay. A delay becomes problematic when the user receives no feedback or when the response takes long enough for the user to think the system is not working. According to Nielsen's usability guidance, a system should provide feedback within about 1 second to keep the user informed. Based on this, the interface should show feedback within 1 second, and AI responses taking longer than approximately 5 seconds are considered slow for this system.

Likelihood: Medium

Impact: Medium to high

Mitigation Strategy:

- Display a loading indicator shortly after the user sends a request.
- Prevent duplicate requests while an AI request is already loading.
- Aim to keep most AI responses within approximately 5 seconds.
- Show an error or fallback message if the AI request fails or takes too long.

R12:Over-Reliance on AI Decision Logic

Description:

Relying too heavily on AI decision-making may reduce predictability and reliability. AI-generated responses may not always match strict safety or operational requirements, especially in a smart home context.

Likelihood: Medium

Impact: Medium

Mitigation Strategy:

- Ensure safety-critical actions are handled outside the AI module.
- Keep control logic within the device or system layer, independent of AI processing

R13: AI Service Dependency and External AI Service Failure

Description:

The system depends on an external AI service. If the service becomes unavailable, unstable, or slow, AI-based command processing may fail or be delayed. From the user's perspective, a response not received within approximately 5–10 seconds may be experienced as a system failure.

Likelihood: Medium

Impact: High

Mitigation Strategy:

- Display a fallback response when the AI service fails or times out.
- Inform the user when network or backend connectivity may be causing the issue.
- Keep normal manual device-control features available even if AI is unavailable.
- Log or observe failures during testing for future improvement.

R14. Incorrect AI response in safety-critical Scenarios

Description:

If the AI gives an incorrect response related to safety situations such as fire, gas leakage, or emergency conditions, it may mislead the user or delay appropriate action. In safety-critical situations, actions must not depend on AI interpretation.

Likelihood: Low to medium

Impact: High

Mitigation Strategy:

- Do not allow AI to directly control emergency responses.
- Ensure fire, gas, and alarm responses are handled by the sensor/device subsystem.
- Use AI only for guidance, explanation, or non-critical user interaction.
- Prioritize automatic safety logic over AI-generated responses.

R15 – Dependency on external speech recognition library

Description:

The voice-control system depends on browser-based speech recognition through the Web Speech API. Browser compatibility differences, microphone permission restrictions, or unsupported browsers may affect reliability or prevent speech recognition from functioning correctly. Speech recognition may also behave inconsistently depending on network connectivity and browser implementation.

Likelihood: Medium

Impact: Medium to High

Mitigation Strategy:

- Test speech recognition functionality on supported browsers such as Chrome and Edge
- Provide visible transcript feedback to confirm recognized speech input
- Automatically stop listening after command execution to reduce unintended repeated commands
- Keep standard manual controls available if speech recognition fails
- Inform users when browser support is unavailable or limited

R16 - Insufficient verification of web application and device-control integration

Description:

There is a risk that the web application appears to function correctly at interface level while the underlying device-control flow behaves incorrectly when connected to Firebase and the

device state backend. This may lead to incorrect updates, inconsistent displayed states, or unnoticed faults in integration between the web interface and the device-control logic.

Probability: Medium

Impact: High

R16 – Browser compatibility and Web Speech API limitations

Description:

The Web Speech API is not supported consistently across all browsers and devices. Differences in implementation or lack of support may affect speech recognition reliability.

Mitigation Strategy:

Test functionality across supported browsers.

Provide fallback to text input where speech is unavailable.

Inform users when speech recognition is not supported.

Probability: Low

Impact :High

Handling Plan:

- Perform manual integration testing on the implemented web application flows
- Verify that frontend actions result in correct Firestore updates
- Verify that Firestore state changes are reflected correctly in the hub interface
- Prioritize testing of login, signup, guest access, hub loading, and toggle actions
- Record defects and perform retesting after fixes
- Consider lightweight unit testing for frontend validation logic in a later step

Updated Handling Status:

Initial automated testing has been implemented for the web application. The current tests verify the authentication and Firebase configuration foundation used by the web application. This improves confidence that the login and signup flows behave correctly and that Firebase Auth and Firestore configuration objects are exported correctly.

However, the complete device-control integration flow is not yet fully automated. Additional tests should be added for the hub page, device toggles, Firestore update calls, and snapshot-based synchronization before this risk can be considered fully managed.

Final Handling Status

The verification risk for web application and device-control integration was reduced significantly. Automated tests now verify hub state display, device-control update calls, Firestore interaction, emergency navigation, and music behavior using mocked services.

The device-control verification is considered managed for the final web prototype scope.

Summary

The SmartHouse project involves technical integration, innovative feature development, and collaborative software engineering. The most critical risks relate to integration complexity, time constraints, and over-ambitious innovation goals. By applying structured planning, modular design, continuous testing, and clear team coordination, these risks can be significantly reduced.

Proper risk management ensures that innovation enhances the project rather than jeopardizes delivery, quality, or maintainability.

Daniel Marcarini - Smart Home code

Risk Description	Priority
R1. Servo power overload and voltage drop	High
R2. Blocking code affecting emergency response	High
R3.Sensor false positives or missed detection	High
R4. State machine logic errors	Medium-High
R5. Hardware component failure	Medium
R6. Electrical wiring instability	Medium

R7. Limited hardware scalability	Medium
R8. Insufficient embedded testing	High

Risk Handling Plans

1 Introduction

This section identifies and evaluates the technical risks associated with the embedded Smart House implementation. Since the system relies on hardware integration, real-time monitoring, and state-based logic, both software and hardware risks must be considered.

Each risk is evaluated in terms of:

- Description
- Likelihood (Low / Medium / High)
- Impact (Low / Medium / High)
- Mitigation Strategy

Technical Risks

Risk 1: Servo Power Overload and Voltage Drop

Description:

Servo motors draw significantly more current than other components. Simultaneous activation of servos and alarm systems may cause voltage instability, unexpected resets, or system malfunction.

Likelihood: Medium

Impact: High

Mitigation Strategy:

- Avoid activating multiple high-current components simultaneously.
- Sequence actuator operations when possible.
- Ensure stable and sufficient power supply.
- Test system behavior under peak load conditions.

Risk 2: Blocking Code Affecting Emergency Response

Description:

If blocking delays are used in the main loop, the system may become temporarily unresponsive. This may delay safety-critical responses such as gas alarm activation.

Likelihood: Medium

Impact: High

Mitigation Strategy:

- Use non-blocking timing mechanisms.
- Avoid delay-based logic in emergency procedures.
- Continuously test responsiveness under multiple simultaneous conditions.

Risk 3: Sensor False Positives or Missed Detection

Description:

Gas and environmental sensors may produce unstable readings due to noise, environmental interference, or improper calibration. This may lead to false alarms or missed hazard detection.

Likelihood: Medium

Impact: High

Mitigation Strategy:

- Calibrate sensors properly.
- Use threshold validation and stable detection logic.
- Implement state flags to prevent repeated triggering.
- Test under various environmental conditions.

Risk 4: State Machine Logic Errors

Description:

Incorrect state transitions may cause repeated emergency activation, inconsistent window states, or incorrect alarm behavior.

Likelihood: Low to Medium

Impact: High

Mitigation Strategy:

- Use clearly defined state variables.
- Validate transitions during testing.
- Document state logic clearly in design documentation.
- Perform scenario-based testing.

Risk 5: Hardware Component Failure

Description:

Failure of a servo motor, sensor, LCD, or buzzer may reduce system functionality or compromise monitoring capabilities.

Likelihood: Medium

Impact: Medium to High

Mitigation Strategy:

- Isolate safety-critical logic from non-critical components.
- Validate hardware components before integration.
- Replace faulty components promptly.

Risk 6: Electrical Wiring Instability

Description:

Loose wiring, poor grounding, or incorrect connections may cause unstable readings or system malfunction.

Likelihood: Medium

Impact: Medium

Mitigation Strategy:

- Use organized wiring layout.
- Secure physical connections.
- Test wiring integrity before deployment.

Risk 7: Limited Hardware Scalability

Description:

The Arduino Uno platform has limited memory and processing power. Adding additional sensors or complex logic may exceed hardware limitations.

Likelihood: Medium

Impact: Medium

Mitigation Strategy:

- Optimize code efficiency.
- Avoid unnecessary memory usage.
- Plan hardware expansion carefully.

Risk 8: Insufficient Embedded Testing

Description:

Embedded systems interact directly with hardware. Insufficient real-world testing may result in unstable or unsafe system behavior.

Likelihood: Medium

Impact: High

Mitigation Strategy:

- Test each subsystem independently.
- Perform full integration testing.
- Simulate emergency conditions during validation.
- Validate behavior under long-running operation.

REFERENCES

1. Nielsen, J. (1993). *Usability Engineering*. Morgan Kaufmann.